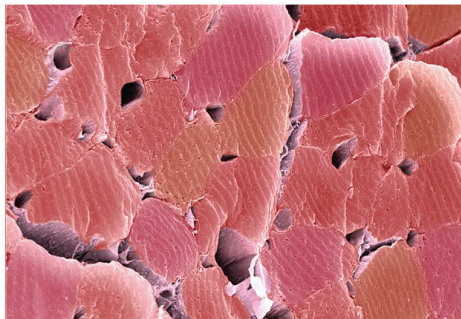


MUSCLES AND TENDONS

MUSCLES CAN ONLY CONTRACT AND SHORTEN. TO RETURN TO THEIR ORIGINAL SHAPE, THEY RELAX AND LENGTHEN PASSIVELY AS OTHER MUSCLES CONTRACT. THE CONTRACTION OF SKELETAL MUSCLES AND TENDONS PRODUCES BODY MOVEMENTS.



STRIATED MUSCLE

This electron micrograph shows a cross-section through skeletal muscle. The bundles of myofibers are interspersed with capillaries (dark areas).

Z band

Where the contractile units (sarcomeres) join end to end

Capillary

M band
Connects neighbouring strands of myosin

MUSCLE STRUCTURE

Skeletal (striated or voluntary) muscle consists of densely packed groups of hugely elongated cells called myofibers. These are grouped into bundles (fascicles). A typical myofibre is $2\text{--}3\text{cm}$ ($\frac{3}{4}\text{--}1\frac{1}{5}\text{in}$) long and 0.05mm ($\frac{1}{500}\text{in}$) in diameter and is composed of narrower structures called myofibrils. These contain thick and thin myofilaments made up mainly of the proteins actin and myosin. Numerous capillaries keep the muscle supplied with the oxygen and glucose needed to fuel contraction.

Thin myofilament
Consists of twisted strands of actin and tropomyosin (protein that inhibits contraction), plus occasional troponin complexes

Actin

Thick myofilament
Main component is the protein myosin; molecules have round heads and long tails

Tropomyosin

Tail of myosin molecule

Head of myosin molecule